

Create hash values

Project description

I used Linux commands to navigate the home directory and examine two files. I compared their plaintext contents, generated SHA-256 hash values for each file using sha256sum, and compared the resulting hashes to identify differences. This demonstrated how cryptographic hashing can be used to verify file integrity and detect changes to data.

Generate hashes for files

I used the ls command to list the contents of the current working directory and the cat command to display all the contents of file1.txt and file2.txt.

```
analyst@adc4d7137a7d:~$ ls
file1.txt  file2.txt
analyst@adc4d7137a7d:~$ cat file1.txt
X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
analyst@adc4d7137a7d:~$ cat file2.txt
X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
9sxa5Yq20Ranalyst@adc4d7137a7d:~$ █
```

I proceeded by using the sha256sum command to generate the hash of the file1.txt file and file2.txt.

```
analyst@adc4d7137a7d:~$ sha256sum file1.txt
131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbfd8267  file1.txt
analyst@adc4d7137a7d:~$ sha256sum file2.txt
2558ba9a4cad1e69804ce03aa2a029526179a91a5e38cb723320e83af9ca017b  file2.txt
analyst@adc4d7137a7d:~$ █
```

Compare hashes

I Used the sha256sum command to generate the hash of the file1.txt file, and send the output to a new file called file1hash then used the same command to generate the hash of the file2.txt file, and send the output to a new file called file2hash. To manually display and compare the differences between the hash values in file1hash and file2hash files I used the cat command.

```
analyst@adc4d7137a7d:~$ sha256sum file1.txt >> file1hash
analyst@adc4d7137a7d:~$ sha256sum file2.txt >> file2hash
analyst@adc4d7137a7d:~$ cat file1hash
131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbfdf8267  file1.txt
131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbfdf8267  file1.txt
analyst@adc4d7137a7d:~$
analyst@adc4d7137a7d:~$ cat file2hash
2558ba9a4cad1e69804ce03aa2a029526179a91a5e38cb723320e83af9ca017b  file2.txt
2558ba9a4cad1e69804ce03aa2a029526179a91a5e38cb723320e83af9ca017b  file2.txt
analyst@adc4d7137a7d:~$ █
```

I then used the `cmp` command to highlight the differences in the `file1hash` and `file2hash` files.

```
analyst@adc4d7137a7d:~$ cmp file1hash file2hash
file1hash file2hash differ: char 1, line 1
analyst@adc4d7137a7d:~$ █
```

The output of the `cmp` command indicates that the hashes differ at the first character in the first line.

Summary

I learned how cryptographic hashing can be used to verify the integrity of files and detect whether data has been changed. I gained hands-on experience using `sha256sum` to generate SHA-256 hashes, comparing hash values, and understanding why even small changes to a file can produce a different hash.